



Institute for
Data Science & Informatics
University of Missouri



HACKATHON

MUIDSI 2026

Agentic **AI** for Real-World Impact

february 13-20, 2026

in partnership
with



agent
platform
sponsor





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Dr. Chi-Ren Shyu

**Associate Dean for Graduate Education and Strategic
Initiatives, College of Engineering
Director, Institute for Data Science and Informatics.
Executive Committee Chair, MU Quantum Innovation Center**



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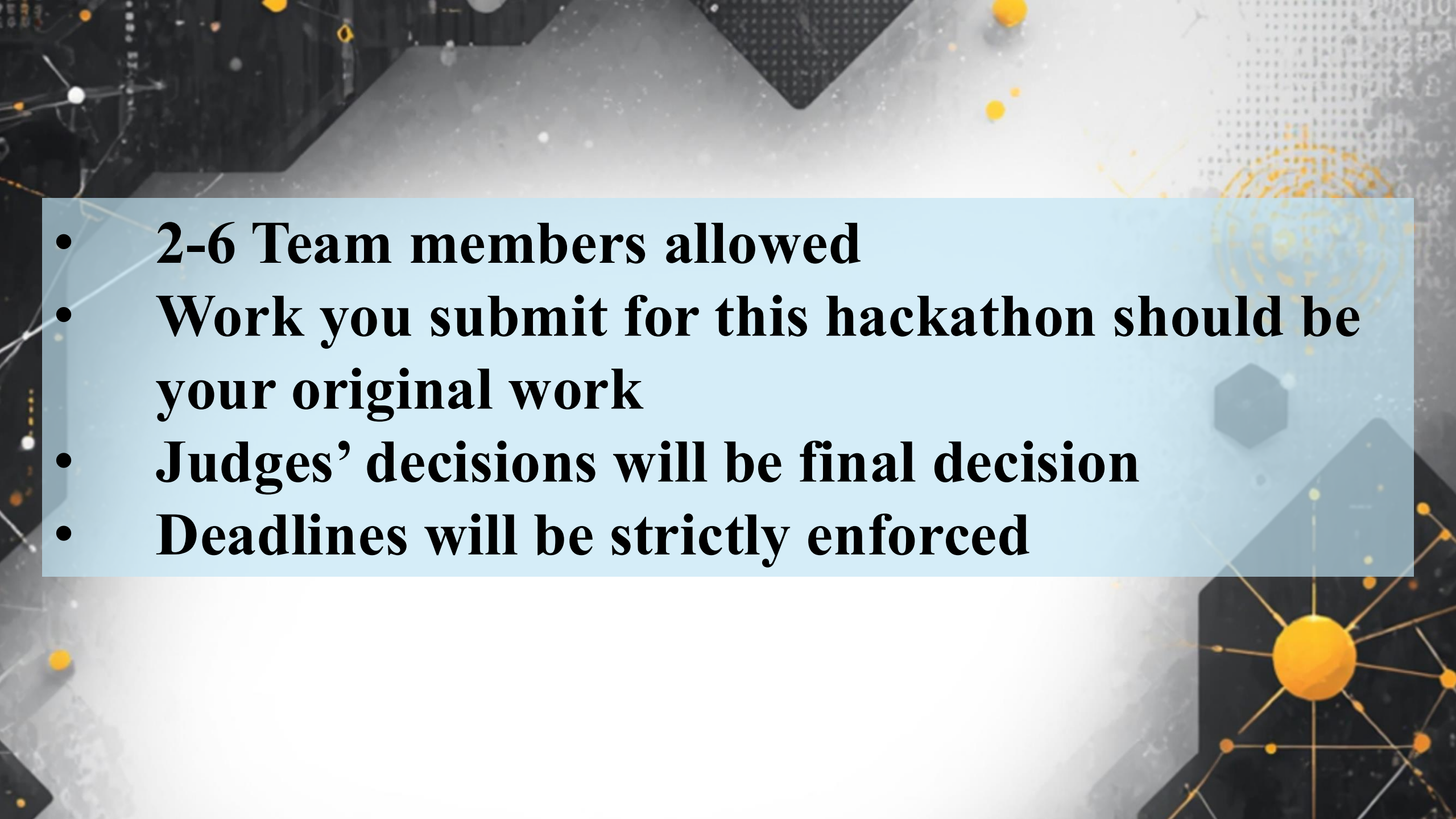


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Rules and Regulations

- 
- The background features a complex, abstract design. It includes a network of nodes and lines, with a prominent yellow sphere at the bottom right. The overall aesthetic is technical and futuristic, with a mix of dark and light tones.
- **2-6 Team members allowed**
 - **Work you submit for this hackathon should be your original work**
 - **Judges' decisions will be final decision**
 - **Deadlines will be strictly enforced**

Tracks

- **Journalism**
- **Nursing**
- **Health/Bio**
- **Geospatial**
- **Agriculture/Plant**
- **Education**
- **Society/Policing/Social**

And Other



Data

Data Provided by our MU Departments

- **Nursing Data**

- This dataset consists of audio data provided by Sinclair School of Nursing
- To access: use the following link, register, and sign data-use agreement
- <https://physionet.org/content/tame-pain/1.0.0/>

- **Journalism Data**

- This dataset consists of geospatial data provided by Reynolds Journalism Institute.
- Go to the Data Access Channel in the 2026 MUIDSI Hackathon Team Chat to find the link.

When using Data Provided by MU Departments

- **Hackathon use only**
- **Please strictly follow the user agreement**
- **Do not share this data with anyone outside your team members**
- **Do not upload Journalism data to cloud platforms even Archia**
- **For Nursing Data Follow instructions provided by <https://physionet.org/>**

* Do the analysis locally when unsure

Geospatial

Climate/Infrastructure

- **USGS**

- <https://www.usgs.gov/products/data/all-data>

- **Climate Data - Geospatial**

- https://catalog.data.gov/dataset/?q=climate+change&sort=views_recent+desc&ext_location=&ext_bbox=&ext_prev_extent=&vocab_category_all=Human+Health&metadata_type=geospatial

- **Climate Data - Federal – Human Health**

- https://catalog.data.gov/dataset/?q=climate&sort=views_recent+desc&organization_type=Federal+Government&ext_location=&ext_bbox=&ext_prev_extent=&vocab_category_all=Human+Health

- **Disaster Data Geospatial**

- https://catalog.data.gov/dataset/?q=disaster&sort=views_recent+desc&metadata_type=geospatial&ext_location=&ext_bbox=&ext_prev_extent=

- **Disaster Data**

- https://catalog.data.gov/dataset/?q=disaster&sort=views_recent+desc&ext_location=&ext_bbox=&ext_prev_extent=&organization_type=Federal+Government

- **Homeland Infrastructure Foundation Level Data (HIFLD)**

- <https://hub.arcgis.com/documents/edc::homeland-infrastructure-foundation-level-data-hifld-open-data/about>
- <https://hifld-geoplatform.hub.arcgis.com/>
- <https://hifld-geoplatform.hub.arcgis.com/pages/hifld-open>

Agriculture/ Soil/Plant

- **Agriculture – Food resilience Geospatial**
 - https://catalog.data.gov/dataset/?q=agriculture&sort=views_recent+desc&metadata_type=geospatial&ext_location=&ext_bbox=&ext_prev_extent=&vocab_category_all=Food+Resilience
- **Agriculture – Food resilience**
 - https://catalog.data.gov/dataset/?q=agriculture&sort=views_recent+desc&ext_location=&ext_bbox=&ext_prev_extent=&vocab_category_all=Food+Resilience

Health/Nursing (Alzheimer's, Autism,Cancer)

- **Alzheimer's Disease Neuroimaging Initiative (ADNI)**
 - <https://adni.loni.usc.edu/data-samples/adni-data/>
- **The National Alzheimer's Coordinating Center (NACC)**
 - <https://naccdata.org/>
- **The Surveillance, Epidemiology, and End Results (SEER)**
 - <https://seer.cancer.gov/data/>
- **The Interactive Autism Network (IAN)**
 - <https://datasetdirectory.disabilitystatistics.org/show/130>
- **Mental Health**
 - <https://www.nimh.nih.gov/health/statistics>
 - https://www.cdc.gov/mentalhealth/data_publications/index.htm
 - <https://catalog.data.gov/dataset?q=mental+health>
 - <https://www.samhsa.gov/data/>
 - <https://www.samhsa.gov/data/report/national-directory-of-mental-health-treatment-facilities>
 - <https://catalog.data.gov/dataset/mental-health-treatment-facilities-locator>
- **Nursing Big Data Repository**
 - https://nursingbigdatarepository.umn.edu/?utm_source=chatgpt.com
- **911 Ambulance Data**
 - <https://nemsis.org/>

Genomic/Bioinformatics

- **TCGA**
 - <https://www.cancer.gov/ccg/research/genome-sequencing/tcga>
- **Single-cell data**
 - <https://www.ncbi.nlm.nih.gov/geo/>
- **Protein database**
 - <https://www.uniprot.org/uniprotkb>

Social/Society

Columbia PD data

- <https://www.como.gov/police/data-reporting-forms/>
- An Additional Columbia PD Traffic Stops Dataset will be shared with you on Teams

More Data

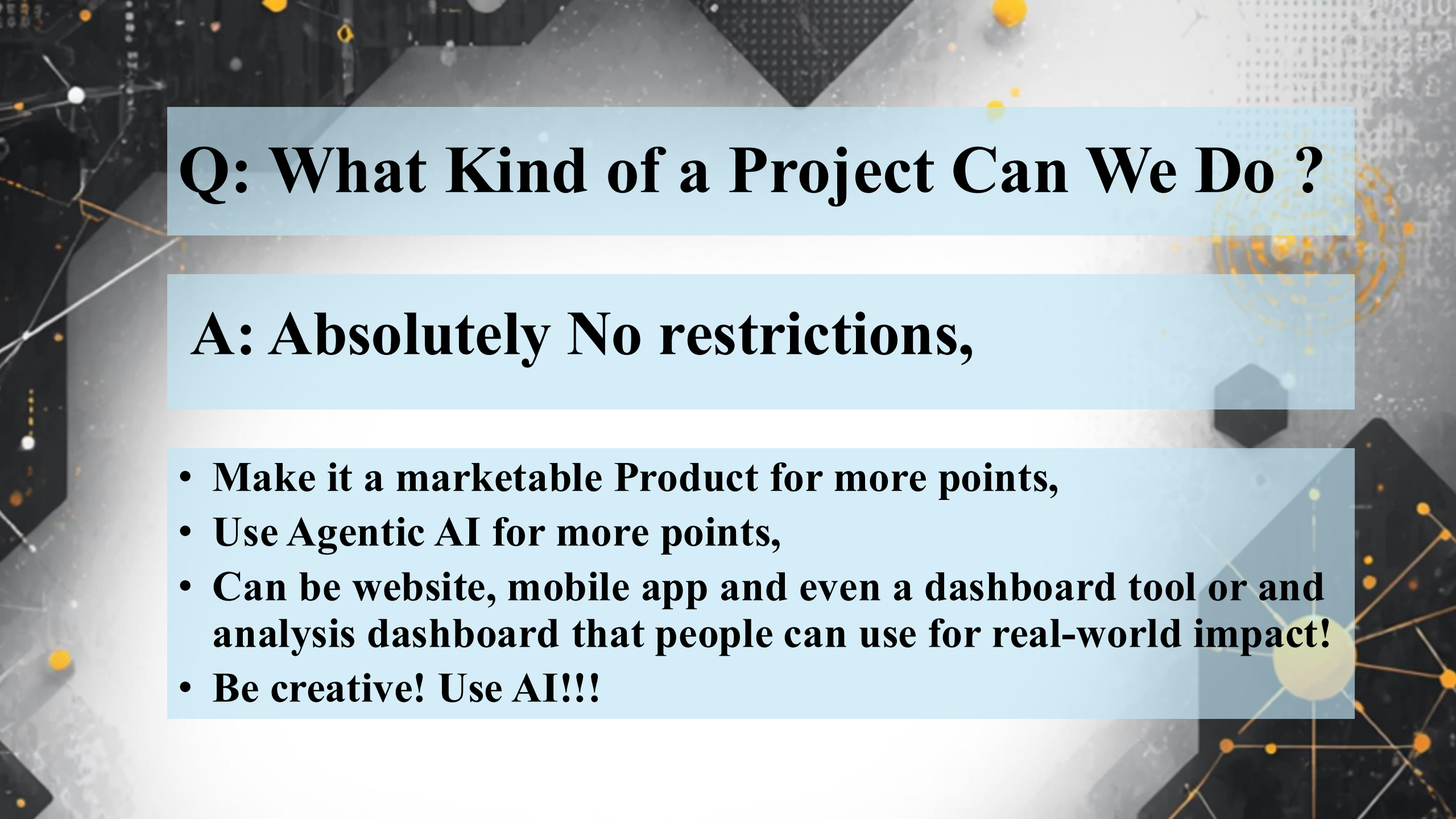
IEEE DataPort

- Large Repository of Data used in ML and AI papers
 - <https://ieee-dataport.org/>

Q: Can I use my own Data ?

A: Of course!

If its public and you have access, no restrictions at all!!

The background of the slide is a complex, abstract composition. It features a dark, textured surface with various geometric shapes, including triangles and hexagons, in shades of grey and black. Scattered throughout are numerous small, glowing yellow and orange dots, some of which are connected by thin, white lines, resembling a network or a molecular structure. A larger, more prominent yellow sphere is visible on the right side, surrounded by smaller dots and lines. The overall effect is one of a futuristic or scientific environment.

Q: What Kind of a Project Can We Do ?

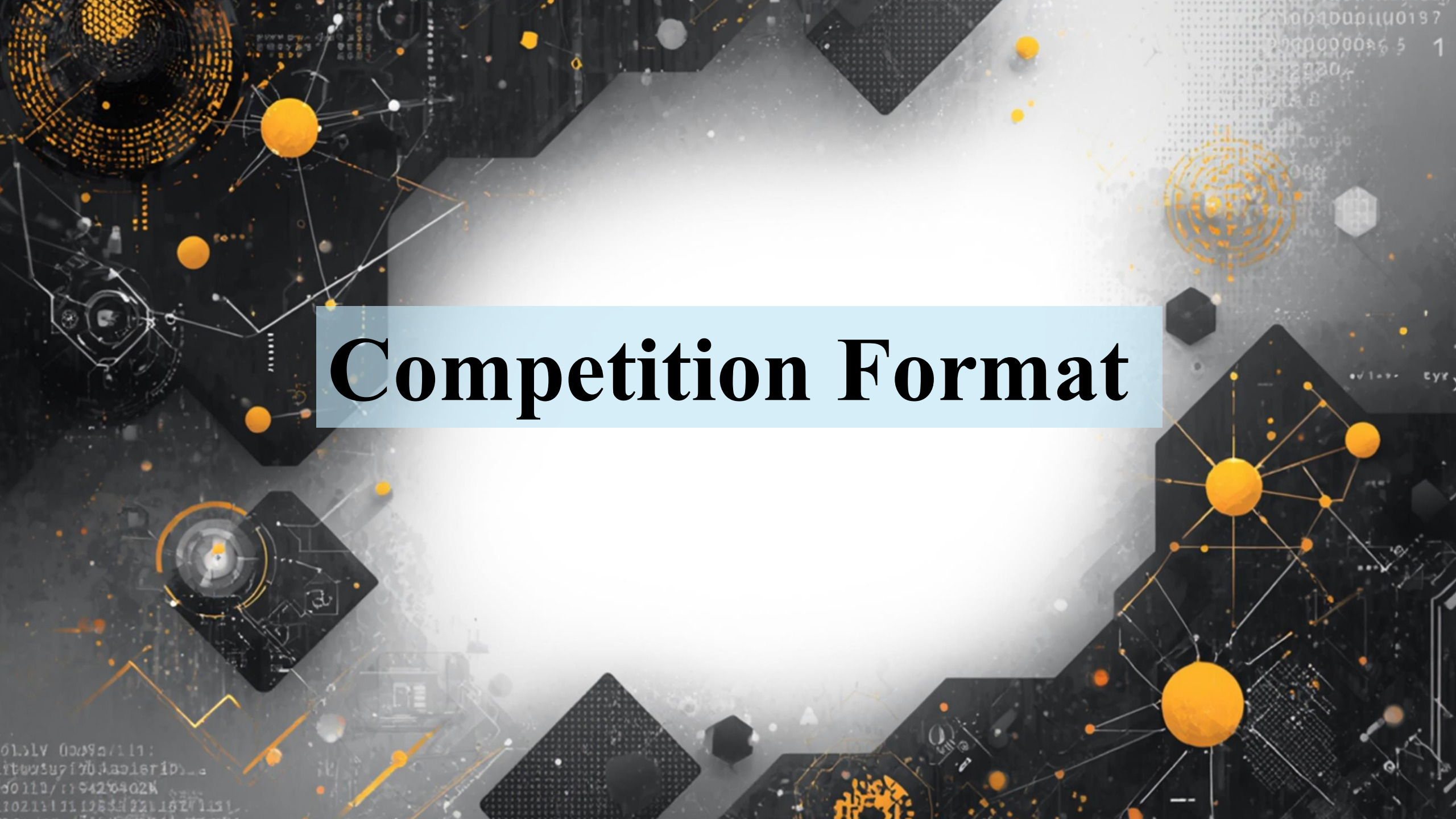
A: Absolutely No restrictions,

- **Make it a marketable Product for more points,**
- **Use Agentic AI for more points,**
- **Can be website, mobile app and even a dashboard tool or and analysis dashboard that people can use for real-world impact!**
- **Be creative! Use AI!!!**

Archia! – For Agentic AI



- **You all have Free tokens in your teams workspace**
- **OpenAI, Gemini, Anthropic.**
- **NOTE: Anthropic models come with a BAA and Zero Data Retention. OpenAI models comes with zero data retention and are generally cheaper than Anthropic.**



Competition Format

Round 1

Kickoff Session

February 13 | 3:30 – 5:30 PM

Reynolds Journalism Institute, Smith Forum Room 200



Work Period

From **February 13** until **February 17, 11:59 PM**, you'll be working on:

- Understanding the data
- Defining the problem you are solving
- Building your application using AI agents
- Grab some snacks, make the weekend fun, and enjoy working with data and AI!

Round 1

Submission – Preliminary Judging

Deadline: February 17 at 11:59 PM (Tuesday Night)

- Each team must submit:
 - A **video presentation (maximum 10 minutes)**
 - Clearly explain:
 - The data you used
 - The problem you're solving
 - Your innovation and excitement
 - Initial analysis, work done, results, and if you have a small demo of your application
 - Remember, the theme is Agentic AI & Archia helps you to build easy AI Agents!
 - There are **no restrictions on creativity**. Submissions will be **offline for the first Round**
- Upload instructions will be shared soon.

Round 1

Judging & Tracks

- Submissions will be evaluated using **RocketJudge**, a digital judging platform.
- Judges include subject-matter experts and industry alumni.
- **First Round Results Announcement**
February 18 (Wednesday) by 8:00 PM
- *Selected teams will receive feedback to improve their projects.*

Round 2 – Finals

February 20 | 1:00 – 5:00 PM

RJI Smith Forum Room 200

10-12 Top teams representing different tracks will present in person and be judged by:

- Industry professionals (including IBM, Archia)
- Faculty members
- We hope to showcase as many teams as possible and crown the **Grand Prize winners!** Plus, and **People's Choice Award!**

(Bring your friends to vote for you and, of course, food!)

Evaluation Criteria

Criteria	Sub-Criteria	Weight (%)
Problem Context & Understanding (10%)	Problem Definition	5
	Social Good Impact	5
Data Exploration & Preparation (30%)	Exploratory Data Analysis (EDA)	10
	Feature Engineering	20
Methodology & Model Development (40%)	Model Development (Selection, Training & Optimization)	30
	Evaluation Metrics	10
Presentation & Academic Communication (10%)	Clarity & Structure	5
	Team Participation & Collaboration	5
Research Impact & Innovation (10%)	Methodological Novelty	10

Use the teams Channel To Reach Us

Questions about teams?

Talk with,

- Lily McEwen

- Samrat Kumar Dey

Questions about Data?

Talk with,

- Joseph Emanuel

Q&A



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Good Luck!
and Have fun!



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